Computer Organization and Assembly Language (COAL)

Lecture 4

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Assembly Language Fundamentals

- **Data Transfer Instructions**
- Addition and Subtraction
- Data-Related Operators and Directives
- Indirect Addressing
- JMP and LOOP Instructions
- 64-Bit Programming

Data Transfer Instructions

- Instruction Syntax
- Operand Types
- Instruction Operand Notation
- Direct Memory Operands
- MOV Instruction
- Zero & Sign Extension
- XCHG Instruction
- Direct-Offset Instructions

Instructions

An assembly instruction contains four basic parts:

- Label (optional)
- Instruction mnemonic (required)
- Operand(s) (usually required)
- Comment (optional)

This is how the different parts are arranged:

```
[label:] mnemonic [operands] [;comment]
```

Operand Types

- **Immediate** – a constant integer (8, 16, or 32 bits)
  - value is encoded within the instruction
- **Register** – the name of a register
  - register name is encoded within the instruction
- **Memory** – reference to a location in memory
  - memory address is encoded within the instruction, or a register holds the address of a memory location

Instruction Operand Notation

Operand	Description
reg8	8-bit general-purpose register: AH, AL, BH, BL, CH, CL, DH, DL
reg16	16-bit general-purpose register: AX, BX, CX, DX, SI, DI, SP, BP
reg32	32-bit general-purpose register: EAX, EBX, ECX, EDX, ESI, EDI, ESP, EBP
reg	Any general-purpose register
sreg	16-bit segment register: CS, DS, SS, ES, FS, GS
imm	8-, 16-, or 32-bit immediate value
imm8	8-bit immediate byte value
imm16	16-bit immediate word value
imm32	32-bit immediate doubleword value
reg/mem8	8-bit operand, which can be an 8-bit general register or memory byte
reg/mem16	16-bit operand, which can be a 16-bit general register or memory word
reg/mem32	32-bit operand, which can be a 32-bit general register or memory doubleword
mem	An 8-, 16-, or 32-bit memory operand

Direct Memory Operands

- A direct memory operand is a named reference to storage in memory
- The named reference (label) is automatically dereferenced by the assembler

```
.data
var1 BYTE 10h
.code
mov al,var1                ; AL = 10h
mov al,[var1]              ; AL = 10h
```

alternate format

Direct Memory Operands

```
.data
var1 BYTE 10h
.code
mov al,var1                    ; AL = 10h
mov al,[var1]                  ; AL = 10h
```

Type Specifier	Bytes addressed
BYTE	1
WORD	2
DWORD	4
QWORD	8
TBYTE	10

MOV Instruction

- Move from source to destination
- Syntax: `MOV` *`destination,source`*
- No more than one memory operand permitted
- Segments, Immediate, EIP, and IP registers cannot be the destination
- Size of operands should be same

```
.data
oneByte BYTE 78h
oneWord WORD 1234h
oneDword DWORD 12345678h

.code
mov  eax,0              ; EAX = 00000000h
```

MOV Instruction

```
.data
   count BYTE 10h
   wVal   WORD 2h
.code

   mov  bl,count
   mov  ax,wVal
   mov  count,al

   mov  al,wVal                    ; error
   mov  ax,count                   ; error
   mov  eax,count                  ; error
```

Your turn . . .

Explain why each of the following MOV statements are invalid:

```
.data
bVal    BYTE    100
bVal2   BYTE    ?
wVal    WORD    2
dVal    DWORD   5
.code

mov ds,45
mov esi,wVal
mov eip,dVal
mov 25,bVal
mov bVal2,bVal
```

Memory – to – Memory

- A single MOV instruction cannot be used to move data directly from one memory location to another
- Move the source operand's value to a register before assigning its value to a memory operand

```
.data
var1 WORD ?
var2 WORD ?
.code
mov  ax,var1
mov  var2,ax
```

Copying Smaller Values to Larger Operands

- Suppose count (unsigned, 16 bits) must be moved to **ECX** (32 bits). We can set ECX to zero and move count to CX:

```
.data
count WORD 1
.code
mov ecx,0
mov cx,count
```

- What happens if we try the same approach with a signed integer equal to -16?

```
.data
signedVal SWORD -16
.code
mov ecx,0
mov cx,signedVal
```

Copying Smaller Values to Larger Operands

- If we had filled ECX first with FFFFFFFFh and then copied signedVal to CX, the final value would have been correct:

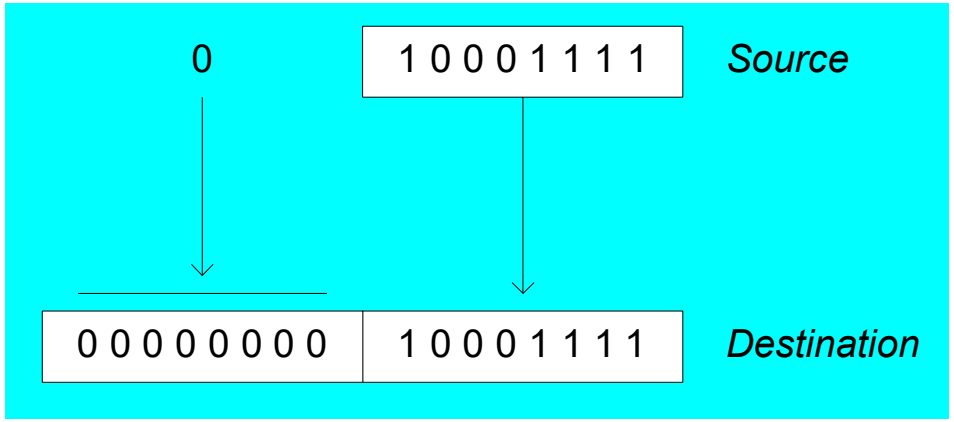
```
mov ecx,0FFFFFFFFh
mov cx,signedVal                ; ECX = FFFFFFF0h (-16)
```

- The effective result of this example was to use the highest bit of the source operand (1) to fill the upper 16 bits of the destination operand, ECX

- This technique is called **sign extension**

Zero Extension - MOVZX

When you copy a smaller value into a larger destination, the MOVZX instruction fills (extends) the upper half of the destination with zeros.




```
mov bl,10001111b
movzx ax,bl                    ; zero-extension
```

Zero Extension - MOVZX

There are three variants:

```
MOVZX   reg32,reg/mem8
MOVZX   reg32,reg/mem16
MOVZX   reg16,reg/mem8
```

- The destination must be a **register.**
- This instruction is only used with **unsigned integers**
- Source operand cannot be a **constant.**
- Source operand needs to be smaller than destination operand

Zero Extension

The following examples use registers for all operands, showing all the size variations:

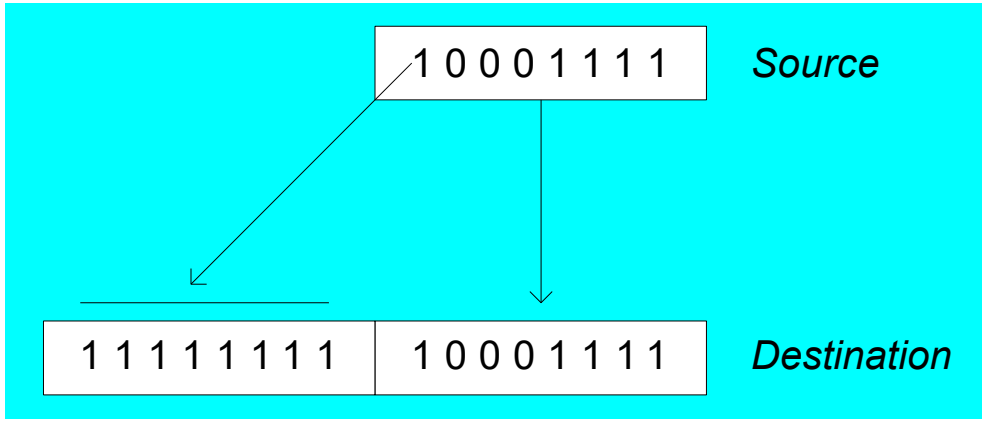
```
mov     bx,0A69Bh
movzx   eax,bx                  ; EAX =
movzx   edx,bl                  ; EDX =
movzx   cx,bl                   ; CX  =
```

The following examples use memory operands for the source and produce the same results:

```
.data
byte1   BYTE 9Bh
word1   WORD 0A69Bh
.code
movzx   eax,word1               ; EAX =
movzx   edx,byte1               ; EDX =
movzx   cx,byte1                ; CX  =
```

Sign Extension

The MOVSX instruction fills the upper half of the destination with a copy of the source operand's sign bit.




```
mov bl,10001111b
movsx ax,bl                    ; sign extension
```

- The destination must be a register.
- This instruction is only used with **signed integers**

LAHF and SAHF

- LAHF = Load status Flags into AH
- Copies the low byte of the EFLAGS register into AH.
- The following flags are copied: **Sign, Zero, Auxiliary Carry** and **Parity**

7	6	5	4	3	2	1	0
SF	ZF	0	AF	0	PF	1	CF

```
.data
saveflags BYTE ?
.code
lahf                        ; load flags into AH
mov  saveflags,ah           ; save them in a variable
```

LAHF and SAHF

- SAHF = Store AH in Status Flags
- Copies AH into the low byte of the EFLAGS (or RFLAGS) register
- You can retrieve the values of flags saved earlier in a variable:

```
mov  ah,saveflags          ; load saved flags into AH
sahf                       ; copy into Flags register
```

XCHG Instruction

- **XCHG** exchanges the values of two operands.
- At least one operand must be a register.
- No immediate operands are permitted.

```
XCHG   reg,reg
XCHG   reg,mem
XCHG   mem,reg
```

```
.data
var1 WORD 1000h
var2 WORD 2000h
.code
xchg ax,bx              ; exchange 16-bit regs
xchg ah,al              ; exchange 8-bit regs
xchg var1,bx            ; exchange mem, reg
xchg eax,ebx            ; exchange 32-bit regs
xchg var1,var2          ; error: two memory operands
```

XCHG Instruction

To exchange two memory operands, use a register as a temporary container and combine MOV with XCHG:

```
mov   ax,val1
xchg  ax,val2
mov   val1,ax
```

Direct-Offset Operands

A constant **offset is added to a data label** to produce an **effective address (EA)**.
The address is dereferenced to get the value inside its memory location.

```
.data
arrayB BYTE 10h,20h,30h,40h
.code
mov al,arrayB+1                ; AL = 20h
mov al,[arrayB+1]              ; alternative notation
```

Q: Why doesn't arrayB+1 produce 11h?

Direct-Offset Operands (cont)

A constant **offset is added to a data label** to produce an **effective address (EA)**.
The address is dereferenced to get the value inside its memory location.

```
.data
arrayW  WORD 1000h,2000h,3000h
arrayD  DWORD 1h,2h,3h,4h
.code
mov ax,[arrayW+2]              ; AX = 2000h
mov ax,[arrayW+4]              ; AX = 3000h
mov eax,[arrayD+4]             ; EAX = 00000002h
```

```
; Will the following statements assemble?
mov ax,[arrayW-2]          ; ??
mov eax,[arrayD+16]        ; ??
```

What will happen when they run?

Your turn. . .

Write a program that rearranges the values of three doubleword values in the following array as: 3, 1, 2.

```
.data
arrayD DWORD 1,2,3
```

- Step1: copy the first value into EAX and exchange it with the value in the second position.

```
mov eax,arrayD
xchg eax,[arrayD+4]
```

- Step 2: Exchange EAX with the third array value and copy the value in EAX to the first array position.

```
xchg eax,[arrayD+8]
mov  arrayD,eax
```

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- **Addition and Subtraction**
- Data-Related Operators and Directives
- Indirect Addressing
- JMP and LOOP Instructions
- 64-Bit Programming

Addition and Subtraction

- INC and DEC Instructions
- ADD and SUB Instructions
- NEG Instruction
- Implementing Arithmetic Expressions
- Flags Affected by Arithmetic

INC and DEC Instructions

- Add 1, subtract 1 from destination operand
  - operand may be **register** or **memory**
- INC *destination*
    - Logic: *destination* ← *destination* + 1
- DEC *destination*
    - Logic: *destination* ← *destination* – 1

- Flags Affected by INC and DEC instructions
  - Zero
  - Sign
  - Overflow
  - Auxiliary

INC and DEC Examples

```
.data
myWord  WORD 1000h
myDword DWORD 10000000h
.code
inc myWord
dec myWord
inc myDword

mov ax,00FFh
inc ax
mov ax,00FFh
inc al
```

ADD and SUB Instructions

- ADD destination, source
  - Logic: *destination* ← *destination* + source
- SUB destination, source
  - Logic: *destination* ← *destination* – source
- Same operand rules as for the MOV instruction
- The ***Carry***, ***Zero***, ***Sign***, ***Overflow***, *and* ***Auxiliary Carry*** are changed according to the value that is placed in the destination operand

ADD and SUB Examples

```
.data
var1 DWORD 10000h
var2 DWORD 20000h
.code                          ; ---EAX---
    mov eax,var1
    add eax,var2
    add ax,FFFFh
    add eax,1
    sub ax,1
```

Evaluate this . . .

- We want to write a program that adds the following three bytes:

```
.data
myBytes BYTE 80h,66h,A5h
```

- What is your evaluation of the following code?

```
mov al,myBytes
add al,[myBytes+1]
add al,[myBytes+2]
```

- What is your evaluation of the following code?

```
mov ax,myBytes
add ax,[myBytes+1]
add ax,[myBytes+2]
```

Evaluate this . . . (cont)

```
.data
myBytes BYTE 80h,66h,A5h
```

- How about the following code?

```
movzx ax,myBytes
mov   bl,[myBytes+1]
add   ax,bx
mov   bl,[myBytes+2]
add   ax,bx                    ; AX = sum
```

Move zero's to BX after the MOVZX instruction.

NEG (negate) Instruction

Reverses the sign of an operand. Operand can be a register or memory operand.

```
.data
valB BYTE -1
valW WORD +32767
.code
    mov al,valB             ;
    neg al                  ;
    neg valW                ;
```

Suppose AX contains **–32,768** and we apply NEG to it. Will the result be valid?

NEG Instruction and the Flags

The processor implements NEG using the following internal operation:

```
SUB 0,operand
```

Implementing Arithmetic Expressions

HLL compilers translate mathematical expressions into assembly language. You can do it also. For example:

```
Rval = -Xval + (Yval - Zval)
```

```
Rval DWORD ?
Xval DWORD 26
Yval DWORD 30
Zval DWORD 40

.code
    mov eax,Xval
    neg eax                 ; EAX = -26
    mov ebx,Yval
    sub ebx,Zval            ; EBX = -10
    add eax,ebx
    mov Rval,eax            ; -36
```

Your turn...

Translate the following expression into assembly language.
Do not permit Xval, Yval, or Zval to be modified:

```
Rval = Xval - (-Yval + Zval)
```

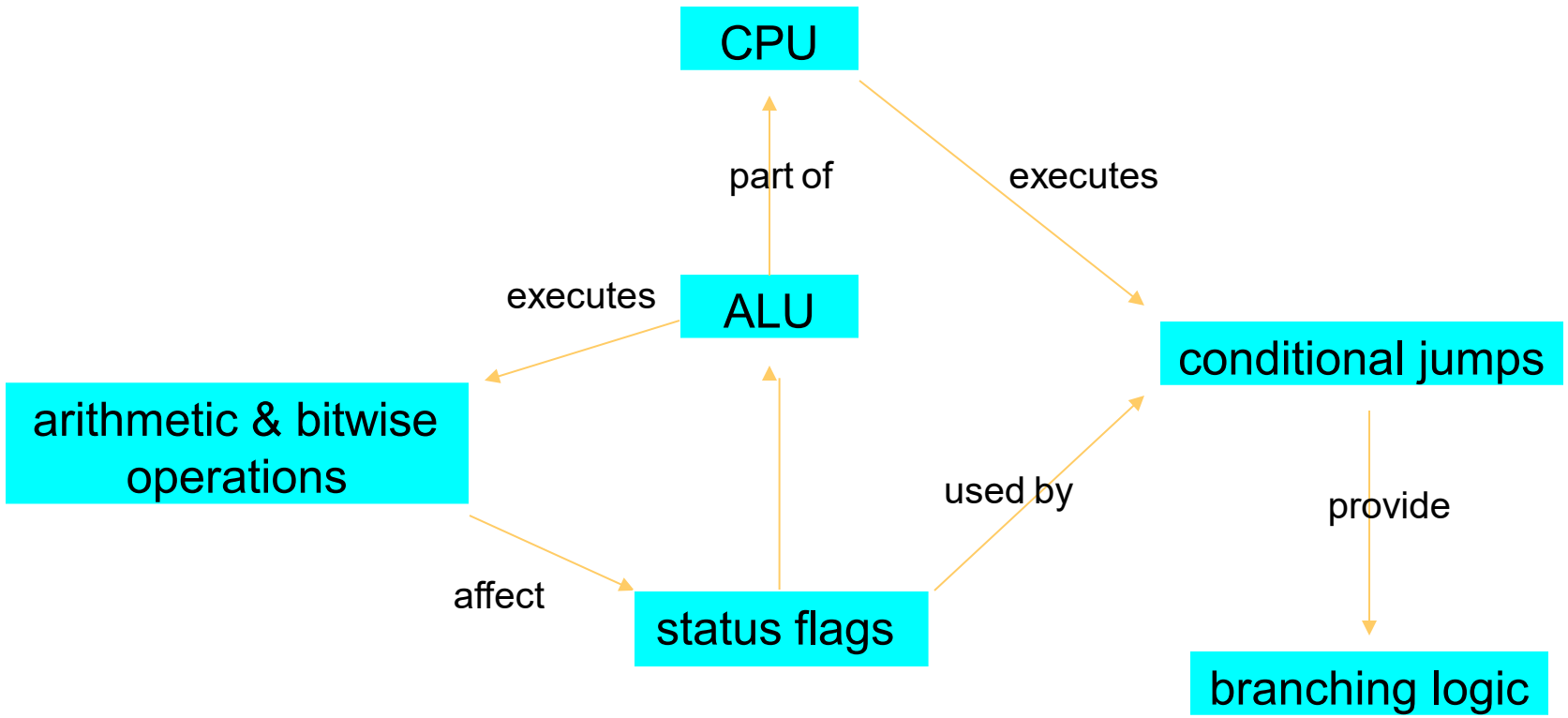
Assume that all values are signed doublewords.

```
mov ebx,Yval
neg ebx
add ebx,Zval
mov eax,Xval
sub eax,ebx
mov Rval,eax
```

Flags Affected by Arithmetic

- The ALU has a number of status flags that reflect the outcome of arithmetic (and bitwise) operations
  - based on the contents of the destination operand
- Essential flags:
  - Zero flag – set when destination equals zero
  - Sign flag – set when destination is negative
  - Carry flag – set when unsigned value is out of range
  - Overflow flag – set when signed value is out of range
- The MOV instruction never affects the flags.

Concept Map



You can use diagrams such as these to express the relationships between assembly language concepts.

Zero Flag (ZF)

The Zero flag is set when the result of an operation produces zero in the destination operand.

```
mov cx,1
sub cx,1            ;
mov ax,0FFFFh
inc ax              ;
inc ax              ;
```

Remember...

- A flag is set when it equals 1.
- A flag is clear when it equals 0.

Sign Flag (SF)

The Sign flag is set when the destination operand is negative.
The flag is clear when the destination is positive.

```
mov cx,0
sub cx,1                    ;
add cx,2                    ;
```

The sign flag is a copy of the destination's highest bit:

```
mov al,0
sub al,1              ;
add al,2              ;
```

Signed and Unsigned Integers: A Hardware Viewpoint

- All CPU instructions operate exactly the same on signed and unsigned integers
- The CPU cannot distinguish between signed and unsigned integers
- **YOU**, the programmer, are solely responsible for using the correct data type with each instruction

Carry Flag (CF)

The Carry flag is set when the result of an operation generates an unsigned value that is out of range (too big or too small for the destination operand).

```
mov al,FFh
add al,1                    ;

; Try to go below zero:

mov al,0
sub al,1                    ;
```

Your turn . . .

For each of the following marked entries, show the values of the destination operand and the Sign, Zero, and Carry flags:

```
mov ax,00FFh
add ax,1            ;
sub ax,1            ;
add al,1            ;
mov bh,6Ch
add bh,95h          ;

mov al,2
sub al,3            ;
```

Overflow Flag (OF)

The Overflow flag is set when the signed result of an operation is invalid or out of range.

```
; Example 1
mov al,+127
add al,1                    ; OF = 1,   AL = ??

; Example 2
mov al,7Fh                  ; OF = 1,   AL = 80h
add al,1
```

The two examples are identical at the binary level because 7Fh equals +127. To determine the value of the destination operand, it is often easier to calculate in hexadecimal.

Your turn . . .

What will be the values of the given flags after each operation?

```
mov al,-128
neg al                ;

mov ax,8000h
add ax,2              ;

mov ax,0
sub ax,2              ;

mov al,-5
sub al,+125           ;
```

What's Next

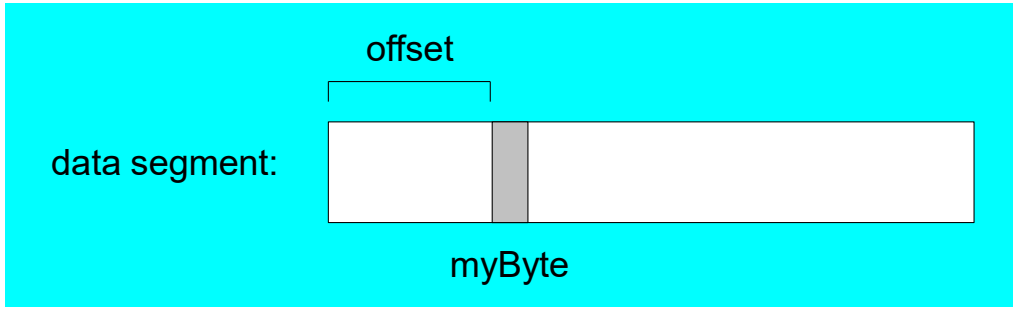
- Data Transfer Instructions
- Addition and Subtraction
- **Data-Related Operators and Directives**
- Indirect Addressing
- JMP and LOOP Instructions
- 64-Bit Programming

Data-Related Operators and Directives

- OFFSET Operator
- PTR Operator
- TYPE Operator
- LENGTHOF Operator
- SIZEOF Operator
- LABEL Directive

OFFSET Operator

- OFFSET returns the address of the label



OFFSET Examples

Let's assume that the data segment begins at 00404000h:

```
.data
bVal BYTE ?
wVal WORD ?
dVal DWORD ?
dVal2 DWORD ?

.code
mov esi,OFFSET bVal          ; ESI = 00404000
mov esi,OFFSET wVal          ; ESI = 00404001
mov esi,OFFSET dVal          ; ESI = 00404003
mov esi,OFFSET dVal2         ; ESI = 00404007
```

Relating to C/C++

The value returned by OFFSET is a pointer. Compare the following code written for both C++ and assembly language:

```
// C++ version:

char array[2];
char * p = array;
```

```
; Assembly language:

.data
array BYTE 01h, C5h
.code
mov esi,OFFSET array
```

ALIGN directive

- aligns a variable on a byte, word or doubleword paragraph boundary
- Syntax: ALIGN *bound*

- Bound can be 1, 2, 4, 8, or 16

In the following example, **bVal** is arbitrarily located at offset 00404000. Inserting the ALIGN 2 directive before **wVal** causes it to be assigned an even-numbered offset:

```
bVal  BYTE  ?                    ; 00404000h
ALIGN 2
wVal  WORD  ?                    ; 00404002h
bVal2 BYTE  ?                    ; 00404004h
ALIGN 4
dVal  DWORD ?                    ; 00404008h
dVal2 DWORD ?                    ; 0040400Ch
```

PTR Operator

Overrides the default type of a label (variable). Provides the flexibility to access part of a variable.

```
.data
myDouble DWORD 12345678h
.code
mov ax,myDouble

mov ax,WORD PTR myDouble

mov WORD PTR myDouble,4321h
```

Little Endian Order

- Little endian order refers to the way Intel stores integers in memory.
- Multi-byte integers are stored in reverse order, with the least significant byte stored at the lowest address
- For example, the doubleword 12345678h would be stored as:

byte	offset
78	0000
56	0001
34	0002
12	0003

When integers are loaded from memory into registers, the bytes are automatically re-reversed into their correct positions.

PTR Operator Examples

```
.data
myDouble DWORD 12345678h
```

doubleword	word	byte	offset	
12345678	5678	78	0000	myDouble
		56	0001	myDouble + 1
		34	0002	myDouble + 2
		12	0003	myDouble + 3

```
mov al,BYTE PTR  myDouble
mov al,BYTE PTR [myDouble+1]
mov al,BYTE PTR [myDouble+2]
mov ax,WORD PTR  myDouble
mov ax,WORD PTR [myDouble+2]
```

PTR Operator (cont)

PTR can also be used to combine elements of a smaller data type and move them into a larger operand. The CPU will automatically reverse the bytes.

```
.data
myBytes BYTE 12h,34h,56h,78h

.code


mov ax,WORD PTR [myBytes]
mov ax,WORD PTR [myBytes+2]
mov eax,DWORD PTR myBytes
```

Your turn . . .

Write down the value of each destination operand:

```
.data
varB BYTE 65h,31h,02h,05h
varW WORD 6543h,1202h
varD DWORD 12345678h

.code
mov ax,WORD PTR [varB+2]
mov bl,BYTE PTR varD
mov bl,BYTE PTR [varW+2]
mov ax,WORD PTR [varD+2]
mov eax,DWORD PTR varW
```

TYPE Operator

The TYPE operator returns the size, in bytes, of a single element of a data declaration.

```
.data
var1 BYTE ?
var2 WORD ?
var3 DWORD ?
var4 QWORD ?

.code

mov eax,TYPE var1          ; 1
mov eax,TYPE var2          ; 2
mov eax,TYPE var3          ; 4
mov eax,TYPE var4          ; 8
```

LENGTHOF Operator

The LENGTHOF operator counts the number of elements in a single data declaration.

```
.data                                   LENGTHOF
byte1  BYTE 10,20,30                    ; 3
array1 WORD 30 DUP(?),0,0               ; 32
array2 WORD 5 DUP(3 DUP(?))             ; 15
array3 DWORD 1,2,3,4                    ; 4
digitStr BYTE "12345678",0              ; 9

.code
mov ecx,LENGTHOF array1                 ; 32
```

SIZEOF Operator

The SIZEOF operator returns a value that is equivalent to multiplying LENGTHOF by TYPE.

```
.data                                   SIZEOF
byte1  BYTE 10,20,30                    ; 3
array1 WORD 30 DUP(?),0,0               ; 64
array2 WORD 5 DUP(3 DUP(?))             ; 30
array3 DWORD 1,2,3,4                    ; 16
digitStr BYTE "12345678",0              ; 9

.code
mov ecx,SIZEOF array1                   ; 64
```

Spanning Multiple Lines (1 of 2)

A data declaration spans multiple lines if each line (except the last) ends with a comma. The LENGTHOF and SIZEOF operators include all lines belonging to the declaration:

```
.data
array WORD 10,20,
    30,40,
    50,60


.code
mov eax,LENGTHOF array        ; 6
mov ebx,SIZEOF array          ; 12
```

Spanning Multiple Lines (2 of 2)

In the following example, array identifies only the first WORD declaration. Compare the values returned by LENGTHOF and SIZEOF here to those in the previous slide:

```
.data
array  WORD 10,20
       WORD 30,40
       WORD 50,60

.code
mov eax,LENGTHOF array     ; 2
mov ebx,SIZEOF array       ; 4
```

LABEL Directive

- Assigns an alternate label name and type to an existing storage location
- LABEL does not allocate any storage of its own
- Removes the need for the PTR operator

```
.data
dwList   LABEL DWORD
wordList LABEL WORD
intList  BYTE 00h,10h,00h,20h
.code

mov eax,dwList              ; 20001000h
mov cx,wordList             ; 1000h
mov dl,intList              ; 00h
```

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Indirect Addressing

- Indirect Operands
- Array Sum Example
- Indexed Operands
- Pointers

Indirect Operands (1 of 2)

An indirect operand holds the address of a variable, usually an array or string. It can be dereferenced (just like a pointer).

```
.data
val1 BYTE 10h,20h,30h
.code
mov esi,OFFSET val1
mov al,[esi]                ; dereference ESI (AL = 10h)

inc esi
mov al,[esi]                ; AL = 20h

inc esi
mov al,[esi]                ; AL = 30h
```

Indirect Operands (2 of 2)

Use PTR to clarify the size attribute of a memory operand.

```
.data
myCount WORD 0

.code
mov esi,OFFSET myCount
inc [esi]                    ; error: ambiguous
inc WORD PTR [esi]           ; ok
```

Should PTR be used here?

```
add [esi],20
```

yes, because [esi] could point to a byte, word, or doubleword

Array Sum Example

Indirect operands are ideal for traversing an array. Note that the register in brackets must be incremented by a value that matches the array type.

```
.data
arrayW WORD 1000h,2000h,3000h
.code
    mov esi,OFFSET arrayW
    mov ax,[esi]
    add esi,2                   ; or: add esi,TYPE arrayW
    add ax,[esi]
    add esi,2
    add ax,[esi]                ; AX = sum of the array
```

ToDo: Modify this example for an array of doublewords.

Indexed Operands

An indexed operand adds a constant to a register to generate an effective address. There are two notational forms:

```
        [label + reg]                    label[reg]

.data
arrayW WORD 1000h,2000h,3000h
.code
    mov esi,0
    mov ax,[arrayW + esi]            ; AX = 1000h
    mov ax,arrayW[esi]               ; alternate format
    add esi,2
    add ax,[arrayW + esi]
    etc.
```

ToDo: Modify this example for an array of doublewords.

Index Scaling

You can scale an indirect or indexed operand to the offset of an array element. This is done by multiplying the index by the array's TYPE:

```
.data
arrayB BYTE  0,1,2,3,4,5
arrayW WORD  0,1,2,3,4,5
arrayD DWORD 0,1,2,3,4,5

.code
mov esi,4
mov al,arrayB[esi*TYPE arrayB]     ; 04
mov bx,arrayW[esi*TYPE arrayW]     ; 0004
mov edx,arrayD[esi*TYPE arrayD]    ; 00000004
```

Pointers

You can declare a pointer variable that contains the offset of another variable.

```
.data
arrayW WORD 1000h,2000h,3000h
ptrW DWORD arrayW
.code
    mov esi,ptrW
    mov ax,[esi]            ; AX = 1000h
```

Alternate format:

```
ptrW DWORD OFFSET arrayW
```

What's Next

- Data Transfer Instructions
- Addition and Subtraction
- Data-Related Operators and Directives
- Indirect Addressing
- **JMP and LOOP Instructions**
- 64-Bit Programming

JMP and LOOP Instructions

- JMP Instruction
- LOOP Instruction
- LOOP Example
- Summing an Integer Array
- Copying a String

JMP Instruction

- JMP is an unconditional jump to a label that is usually within the same procedure.
- Syntax: JMP *target*
- Logic: EIP ← *target*
- Example:

```
top:
    .
    .
    jmp top
```

A jump outside the current procedure must be to a special type of label called a global label (see Section 5.5.2.3 for details).

LOOP Instruction

- The LOOP instruction creates a counting loop
- Syntax: LOOP *target*
- Logic:
  - ECX ← ECX – 1
  - if ECX != 0, jump to *target*
- Implementation:
  - The assembler calculates the distance, in bytes, between the offset of the following instruction and the offset of the target label. It is called the relative offset.
  - The relative offset is added to EIP.

LOOP Example

The following loop calculates the sum of the integers 5 + 4 + 3 +2 + 1:

```
offset      machine code          source code
00000000    66 B8 0000            mov   ax,0
00000004    B9 00000005           mov   ecx,5

00000009    66 03 C1          L1: add   ax,cx
0000000C    E2 FB                 loop L1
0000000E
```

When LOOP is assembled, the current location = 0000000E (offset of the next instruction). –5 (FBh) is added to the the current location, causing a jump to location 00000009:

00000009 ← 0000000E + FB

Your turn . . .

If the relative offset is encoded in a single signed byte,

(a) what is the largest possible backward jump?

(b) what is the largest possible forward jump?

(a) –128

(b) +127

Your turn . . .

What will be the final value of AX?

10

```
    mov ax,6
    mov ecx,4
L1:
    inc ax
    loop L1
```

How many times will the loop execute?

4,294,967,296

```
    mov ecx,0
X2:
    inc ax
    loop X2
```

Nested Loop

If you need to code a loop within a loop, you must save the outer loop counter's ECX value. In the following example, the outer loop executes 100 times, and the inner loop 20 times.

```
.data
count DWORD ?
.code
    mov ecx,100             ; set outer loop count
L1:
    mov count,ecx           ; save outer loop count
    mov ecx,20              ; set inner loop count
L2: .
    .
    loop L2                 ; repeat the inner loop
    mov ecx,count           ; restore outer loop count
    loop L1                 ; repeat the outer loop
```

What's Next

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- **64-Bit Programming**

64-Bit Programming

- MOV instruction in 64-bit mode accepts operands of 8, 16, 32, or 64 bits
- When you move a 8, 16, or 32-bit constant to a 64-bit register, the upper bits of the destination are cleared.
- When you move a memory operand into a 64-bit register, the results vary:
  - 32-bit move clears high bits in destination
  - 8-bit or 16-bit move does not affect high bits in destination

More 64-Bit Programming

- MOVSXD sign extends a 32-bit value into a 64-bit destination register
- The OFFSET operator generates a 64-bit address
- LOOP uses the 64-bit RCX register as a counter
- RSI and RDI are the most common 64-bit index registers for accessing arrays.

Other 64-Bit Notes

- ADD and SUB affect the flags in the same way as in 32-bit mode
- You can use scale factors with indexed operands.

Summary

- Data Transfer
  - MOV – data transfer from source to destination
  - MOVSX, MOVZX, XCHG
- Operand types
  - direct, direct-offset, indirect, indexed
- Arithmetic
  - INC, DEC, ADD, SUB, NEG
  - Sign, Carry, Zero, Overflow flags
- Operators
  - OFFSET, PTR, TYPE, LENGTHOF, SIZEOF, TYPEDEF
- JMP and LOOP – branching instructions

Thanks a lot

If you are taking a Nap, **wake up**.....Lecture Over